https://github.com/shrihanhegdebca25-blip/DBMS-4-6

```
CREATE TABLE STUDENT (
    Student_ID INT,
    Student_Name VARCHAR(50),
    Class VARCHAR(100),
    Marks INT,
    Fee_Paid DECIMAL(10,2),
    Date_of_Birth DATE,
    Admission_Date DATE
);

INSERT INTO STUDENT
(Student_ID, Student_Name, Class, Marks, Fee_Paid, Date_of_Birth,
Admission_Date)
VALUES
(101, 'Shrihan', '10A', 85, 50000.00, '2007-06-05', '2023-06-10'),
(102, 'Shourya', '10A', 92, 50000.00, '2008-11-20', '2023-06-12'),
(103, 'Sachith', '10B', 76, 45000.00, '2009-02-10', '2023-06-11'),
(104, 'Raghu',   '10B', 88, 48000.00, '2008-08-25', '2023-06-15'),
(105, 'Mohith',  '10A', 65, 40000.00, '2009-06-05', '2023-06-13');
```

4. Implement SQL Functions: Conversion Functions, and Date Functions.

```
SELECT CAST('2025' AS UNSIGNED) + 5 AS Cast_Result;

SELECT CONVERT('123', SIGNED) + 7 AS Convert_Result;

SELECT Student_Name, CAST(Fee_Paid AS SIGNED) AS Fee_Integer
FROM STUDENT;

SELECT Student_Name, CAST(Marks AS CHAR) AS Marks_Text
FROM STUDENT;

SELECT STR_TO_DATE('15-08-2024', '%d-%m-%Y') AS Converted_Date;

SELECT Student_Name, DATE_FORMAT(Admission_Date, '%d %b %Y') AS
Admission_Formatted
FROM STUDENT;
```

```sql
SELECT Student_Name, FORMAT(Fee_Paid, 2) AS Fee_Formatted
FROM STUDENT;

-- Date functions
SELECT Student_Name,
       YEAR(Date_of_Birth)  AS Birth_Year,
       MONTH(Date_of_Birth) AS Birth_Month,
       DAY(Date_of_Birth)   AS Birth_Day,
       DAYNAME(Date_of_Birth) AS Birth_Weekday
FROM STUDENT;

SELECT Student_Name,
       Admission_Date,
       DATE_ADD(Admission_Date, INTERVAL 1 YEAR)  AS One_Year_Later,
       DATE_SUB(Admission_Date, INTERVAL 15 DAY)  AS Fifteen_Days_Before
FROM STUDENT;

SELECT Student_Name,
       DATEDIFF(CURDATE(), Admission_Date) AS Days_Since_Admission,
       TIMESTAMPDIFF(YEAR, Date_of_Birth, CURDATE()) AS Age
FROM STUDENT;

SELECT CURDATE() AS Today, LAST_DAY('2024-02-10') AS Month_End;
```

Output:

Student_Name	Admission_Date	One_Year_Later	Fifteen_Days_Before
Shrihan	2023-06-10	2024-06-10	2023-05-26
Shourya	2023-06-12	2024-06-12	2023-05-28
Sachith	2023-06-11	2024-06-11	2023-05-27
Raghu	2023-06-15	2024-06-15	2023-05-31
Mohith	2023-06-13	2024-06-13	2023-05-29

Student_Name	Days_Since_Admission	Age
Shrihan	1210	19
Shourya	1208	17
Sachith	1209	17
Raghu	1205	18
Mohith	1207	17

Today	Month_End
2026-10-02	2024-02-29

5. **Implement SQL Operators: Demonstrate the use of different types of operators in SQL with examples, including Arithmetic Operators, Logical Operators, Comparison Operators.**

```
SELECT 15 + 4 AS Addition, 15 - 4 AS Subtraction, 15 * 4 AS
Multiplication,
       15 / 4 AS Division, 15 DIV 4 AS Int_Division, 15 % 4 AS Modulus;

SELECT Student_Name, Marks,
       Marks + 5 AS Marks_Plus_5,
       Marks * 2 AS Marks_Double,
       Fee_Paid / 12 AS Monthly_Fee,
       Marks % 10 AS Marks_Mod_10
FROM STUDENT;

-- Comparison
SELECT Student_Name, Marks FROM STUDENT WHERE Marks > 80;
SELECT Student_Name, Marks FROM STUDENT WHERE Marks < 76;
SELECT Student_Name, Marks FROM STUDENT WHERE Marks >= 88;
```

```
SELECT Student_Name, Marks FROM STUDENT WHERE Marks <= 76;
SELECT Student_Name, Class FROM STUDENT WHERE Class = '10A';
SELECT Student_Name, Class FROM STUDENT WHERE Class <> '10A';

SELECT 5 = 5 AS Equal, 5 <> 3 AS Not_Equal, 5 > 8 AS Greater,
       5 <=> 5 AS Null_Safe_1, NULL = NULL AS Null_Eq, NULL <=> NULL AS
Null_Safe_2;

-- Logical
SELECT Student_Name, Class, Marks
FROM STUDENT
WHERE Class = '10A' AND Marks > 80;

SELECT Student_Name, Class, Marks
FROM STUDENT
WHERE Class = '10B' OR Marks > 90;

SELECT Student_Name, Marks
FROM STUDENT
WHERE NOT (Marks >= 76);

SELECT Student_Name, Class, Marks
FROM STUDENT
WHERE (Class = '10A') XOR (Marks > 80);
```

6. **Implement SQL Operators: Demonstrate the use of different types of operators in SQL with examples Special Operators, and Set Operations(UNION)**

```
CREATE TABLE Sports_Club (Student_Name VARCHAR(50), Class VARCHAR(100));
CREATE TABLE Music_Club  (Student_Name VARCHAR(50), Class VARCHAR(100));

INSERT INTO Sports_Club VALUES ('Shrihan','10A'), ('Raghu','10B'),
('Mohith','10A');
INSERT INTO Music_Club  VALUES ('Shourya','10A'), ('Raghu','10B'),
('Sachith','10B');

-- Special operators
SELECT Student_Name, Marks FROM STUDENT WHERE Marks BETWEEN 70 AND 90;

SELECT Student_Name, Class FROM STUDENT WHERE Class IN ('10A');
```

```sql
SELECT Student_Name, Marks FROM STUDENT WHERE Marks NOT IN (65, 92);

SELECT Student_Name FROM STUDENT WHERE Student_Name LIKE 'S%';

SELECT Student_Name FROM STUDENT WHERE Student_Name LIKE '_h%';

SELECT Student_Name FROM STUDENT WHERE Student_Name LIKE '%th';

SELECT Student_Name, Fee_Paid FROM STUDENT WHERE Fee_Paid IS NOT NULL;

SELECT Student_Name, Marks
FROM STUDENT
WHERE Marks > ALL (SELECT Marks FROM STUDENT WHERE Class = '10B');

SELECT Student_Name, Marks
FROM STUDENT
WHERE Marks > ANY (SELECT Marks FROM STUDENT WHERE Class = '10B');

-- Set operations
SELECT Student_Name, Class FROM Sports_Club
UNION
SELECT Student_Name, Class FROM Music_Club;

SELECT Student_Name, Class FROM Sports_Club
UNION ALL
SELECT Student_Name, Class FROM Music_Club;
```

```
+----------------+
| Converted_Date |
+----------------+
| 2024-08-15     |
+----------------+
```

Student_Name	Admission_Formatted
Shrihan	10 Jun 2023
Shourya	12 Jun 2023
Sachith	11 Jun 2023
Raghu	15 Jun 2023
Mohith	13 Jun 2023

Student_Name	Fee_Formatted
Shrihan	50,000.00
Shourya	50,000.00
Sachith	45,000.00
Raghu	48,000.00
Mohith	40,000.00

Output:

Student_Name	Birth_Year	Birth_Month	Birth_Day	Birth_Weekday
Shrihan	2007	6	5	Tuesday
Shourya	2008	11	20	Thursday
Sachith	2009	2	10	Tuesday
Raghu	2008	8	25	Monday
Mohith	2009	6	5	Friday

Student_Name	Admission_Date	One_Year_Later	Fifteen_Days_Before
Shrihan	2023-06-10	2024-06-10	2023-05-26
Shourya	2023-06-12	2024-06-12	2023-05-28
Sachith	2023-06-11	2024-06-11	2023-05-27
Raghu	2023-06-15	2024-06-15	2023-05-31
Mohith	2023-06-13	2024-06-13	2023-05-29

Student_Name	Days_Since_Admission	Age
Shrihan	1210	19
Shourya	1208	17
Sachith	1209	17

Today	Month_End
2026-10-02	2024-02-29

Addition	Subtraction	Multiplication	Division	Int_Division	Modulus
19	11	60	3.7500	3	3

Student_Name	Marks	Marks_Plus_5	Marks_Double	Monthly_Fee	Marks_Mod_10
Shrihan	85	90	170	4166.666667	5
Shourya	92	97	184	4166.666667	2
Sachith	76	81	152	3750.000000	6
Raghu	88	93	176	4000.000000	8
Mohith	65	70	130	3333.333333	5

Student_Name	Marks
Shrihan	85
Shourya	92

```
+--------------+-------+
| Student_Name | Marks |
+--------------+-------+
| Mohith       |    65 |
+--------------+-------+
+--------------+-------+
| Student_Name | Marks |
+--------------+-------+
| Shourya      |    92 |
| Raghu        |    88 |
+--------------+-------+
+--------------+-------+
| Student_Name | Marks |
+--------------+-------+
| Sachith      |    76 |
| Mohith       |    65 |
+--------------+-------+
+--------------+-------+
| Student_Name | Class |
+--------------+-------+
| Shrihan      | 10A   |
| Shourya      | 10A   |
| Mohith       | 10A   |
+--------------+-------+
```

Output:

```
| Shrihan      |    85 |
| Shourya      |    92 |
| Raghu        |    88 |
+--------------+-------+
```

```
+--------------+-------+
| Student_Name | Class |
+--------------+-------+
| Shrihan      | 10A   |
| Raghu        | 10B   |
| Mohith       | 10A   |
| Shourya      | 10A   |
| Sachith      | 10B   |
+--------------+-------+
```

```
+--------------+-------+
| Student_Name | Class |
+--------------+-------+
| Shrihan      | 10A   |
| Raghu        | 10B   |
| Mohith       | 10A   |
| Shourya      | 10A   |
| Raghu        | 10B   |
| Sachith      | 10B   |
+--------------+-------+
```